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Optimización del Proceso de Gestión de Talento Humano Docente: Automatización del Ciclo de Vida de Contratación en Universidades Públicas

***Abstract*—The rapid digital transformation of Higher Education Institutions (HEIs) necessitates structural modernization across human resource management, faculty staffing, and administrative workflows. This paper examines contemporary frameworks for academic talent management, evaluating the integration of automated solutions, competency frameworks, and media-driven technological platforms in modern university administration.**

***Index Terms*—Talent Management, Higher Education Institutions, Digitalization, Academic Workflows, Competency Evaluation.**

I. Introducción

The operational landscape of Higher Education Institutions (HEIs) is undergoing a profound structural transition driven by globalized academic competition and technological evolution. As universities expand their administrative and educational scope, the recruitment, integration, and retention of qualified academic personnel have emerged as vital strategic imperatives. Traditional administrative methodologies often fail to address the complexities of modern faculty allocation, necessitating dynamic systems capable of coordinating academic human resources effectively.

Human Resource Management (HRM) within the higher education sector faces distinct challenges compared to corporate environments. The dual obligations of pedagogical excellence and high-impact research require university administration to establish comprehensive evaluation and onboarding mechanisms. Without structured frameworks, academic institutions suffer from inefficiencies in hiring, sub-optimal faculty placement, and high attrition rates, ultimately undermining institutional performance and reputation.

Digitalization plays a pivotal role in overcoming these organizational bottlenecks. The implementation of modern digital platforms, Enterprise Resource Planning (ERP) systems, and automated Applicant Tracking Systems (ATS) enables seamless communication and administrative tracking across university faculties. Digitally integrated administrative environments facilitate the systematic evaluation of candidate competencies while lowering overall administrative overhead.

Furthermore, internationalization strategies have introduced new dimensions to faculty recruitment. Modern HEIs actively recruit global academic talent to boost research output and global ranking indices. However, attracting international faculty members requires robust institutional infrastructure, as the integration process is heavily mediated by the institution's overall level of administrative and operational digitization [**?**].

In parallel, the profiles and core competencies expected of university educators are shifting rapidly in response to the digital age. Pedagogical methodologies increasingly incorporate specialized digital tools and interactive media formats, requiring continuous professional development and training. Consequently, institutional evaluation models must adapt to assess both traditional academic competencies and digital literacy indicators [**?**].

Continuous professional training and the deployment of media education technologies are critical for preparing educators to navigate complex learning environments [**?**]. Integrating modern technological tools into faculty development programs enhances pedagogical effectiveness and aligns academic output with modern industry standards and societal demands.

Despite these technological advancements, many HEIs struggle to operationalize holistic talent management frameworks that seamlessly integrate recruitment, retention, and performance evaluation. Developing structured, sector-specific frameworks is essential to align administrative HR processes with broader institutional goals and quality assurance metrics [**?**].

This review provides a comprehensive analysis of contemporary approaches to academic talent management. By analyzing the intersection of institutional digitization, competency frameworks, media technologies, and talent management architectures, this study outlines actionable insights for optimizing HR workflows in modern higher education environments.

II. Background

To properly contextualize the application of digital platforms and quality management in higher education HR workflows, several fundamental concepts and theoretical frameworks must be established. This section outlines the core principles governing talent management, digital maturity in academia, and competency evaluation.

A. Talent Management Architectures in HEIs

Talent management in higher education encompasses the strategic acquisition, development, and retention of academic and administrative staff. Unlike corporate HR models, university talent architectures must account for tenure systems, academic freedom, research productivity, and teaching effectiveness. Structured frameworks provide operational roadmaps for talent identification, succession planning, and performance

management, ensuring institutional continuity and competitive standing [**?**].

B. Institutional Digitization and Academic Integration

Institutional digitization refers to the extent to which an organization integrates digital technologies across its core operational, communication, and management workflows. In HEIs, high digitization levels streamline administrative procedures, facilitate cross-border faculty collaboration, and reduce friction during international faculty integration [**?**]. Systems such as automated workflows, digital signature verification, and centralized databases form the backbone of modern academic infrastructure.

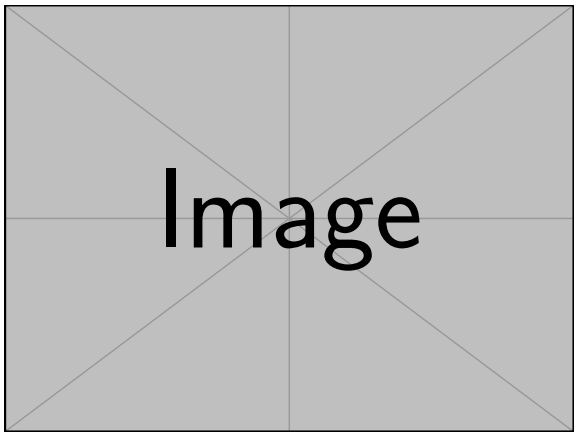


Fig. 1. Conceptual Architecture of Digitalized Talent Management and Quality Assurance in Higher Education Institutions.

C. Competency Frameworks and Media Literacy

The transition toward digitalized learning environments demands clear competency indicators for academic staff. Competency models in the digital age evaluate pedagogical skills alongside technological adaptability and data literacy [**?**]. Furthermore, integrating media education technologies into continuous professional training enables educators to utilize advanced digital communication tools effectively [**?**].

III. Indicadores Clave de Desempeño (KPI)

La optimización se evalúa sobre cinco indicadores, definidos antes de la implementación y medidos de forma continua. La Table I resume su definición y la correspondencia con los módulos de automatización de **??**.

K1 (Lead Time de Contratación) es el indicador agregado del proceso completo. K2 (Tasa de Idoneidad del Perfil) se calcula sobre el total de contratos formalizados por convocatoria. K3 (Costo Administrativo por Contratación) permite comparar el proceso manual contra el automatizado y justificar económicamente la inversión. K4 (Índice de Digitalización de Expedientes) mide el avance real de la transformación. K5 (Tasa de Rotación Temprana) es un indicador de resultado tardío pero crítico: conecta la calidad del proceso de contratación con la continuidad del servicio académico.

TABLE I
Indicadores clave de desempeño: definición, unidad y módulo que actúa sobre cada uno.

KPI	**Definición**	**Módulo**
Lead Time de Contratación	Días desde la detección de la vacante hasta firma del contrato y alta en el sistema	Workflow
Tasa de Idoneidad del Perfil	% de contratados que cumplen al 100% los requisitos técnicos y académicos	Validación
Costo Adm. por Contratación	Horas-hombre y recursos materiales invertidos en la selección de un docente	ATS
Índice de Digitalización de Expedientes	% de procesos gestionados íntegramente de forma digital vs. manual/papel	Workflows
Tasa de Rotación Temprana	% de docentes que abandonan antes del primer año por fallos de selección o inducción	Onboarding